

Answers: Logic Drill #5

Sheet by David Akinyele-Aje

Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

Section A

1. Name the fallacy: “If n is divisible by 6, then n is divisible by 3. $n = 15$ is divisible by 3. Therefore $n = 15$ is divisible by 6.”

Answer: Affirming the consequent

Method: The argument has the form “ $P \implies Q$, Q , therefore P ”, which is invalid: knowing Q is true does not establish P . Indeed 15 is divisible by 3 but not by 6.

Investigate further: Write the valid rule that this fallacy is often confused with (modus ponens).

2. Name the fallacy: “If $x > 10$, then $x > 0$. Suppose $x \leq 10$. Therefore $x \leq 0$.”

Answer: Denying the antecedent

Method: The argument has the form “ $P \implies Q$, $\neg P$, therefore $\neg Q$ ”, which is invalid. E.g. $x = 5$ satisfies $x \leq 10$ but not $x \leq 0$.

Investigate further: What is the valid rule that instead concludes from $\neg Q$ to $\neg P$?

3. True or false: “If a proof of $P \implies Q$ is shown to be flawed, then $P \implies Q$ must be false.”

Answer: False

Method: A flawed proof only invalidates that particular argument; $P \implies Q$ may still be true and simply require a different, correct proof.

Investigate further: Give an example of a true statement that is easy to state but whose standard proof most students get wrong on a first attempt.

4. A 6-line “proof” is flawed only on line 4. How many of the six lines can you actually trust?

Answer: 3 (lines 1–3)

Method: Once line 4 is wrong, every subsequent line (5 and 6) depends on it and cannot be trusted either, even if the algebra within them is locally correct.

Investigate further: Could line 6 coincidentally still be a true statement despite depending on a flawed line 4?

5. True or false: “Squaring both sides of an equation is a reversible step that can never introduce a new, extraneous solution.”

Answer: False

Method: Squaring is not injective on negative numbers: e.g. $x = -2$ does not satisfy $x = 2$, but squaring both sides gives $x^2 = 4$, which is also satisfied by $x = 2$.

Investigate further: Name another common “operation” on an equation that can introduce extraneous solutions.

6. A “proof” concludes $0 = 1$. What does this tell you about the argument (assuming it is not a contradiction proof)?

Answer: At least one line of the argument contains an error.

Method: $0 = 1$ is false. A valid argument cannot derive a false conclusion from true premises, so an error must exist somewhere in the chain of reasoning.

Investigate further: How would your answer change if this occurred inside a proof by contradiction?

7. True or false: “Verifying a universal claim $P(n)$ for $n = 1, 2, \dots, 10$ constitutes a full proof that $P(n)$ holds for all positive integers n .”

Answer: False

Method: Finite verification never proves a $\forall n$ claim; a counterexample could exist at $n = 11$ or beyond.

Investigate further: What proof technique specifically is designed to close this gap for claims about all positive integers?

8. What is the flaw in dividing both sides of $ax = bx$ by x to conclude $a = b$?

Answer: The step is invalid if $x = 0$.

Method: If $x = 0$, $ax = bx$ becomes $0 = 0$, which is true for any a, b — so $a = b$ need not hold. Dividing by a possibly-zero quantity can lose solutions/cases.

Investigate further: What should be written instead to handle the $x = 0$ case correctly?

9. True or false: “In a proof by contradiction, the final derived statement must be a strict logical contradiction (such as $0 = 1$), not merely an unusual-looking result.”

Answer: True

Method: A valid contradiction proof requires deriving something that is genuinely impossible (a statement and its negation both holding, or a false arithmetic identity) — an “unusual” but not actually impossible result does not complete the proof.

Investigate further: Give an example of a result that looks surprising but is not actually a contradiction.

10. What is wrong with concluding from $\frac{a}{b} = \frac{c}{d}$ that $a = c$ and $b = d$?

Answer: Equal fractions do not require equal numerators and denominators separately.

Method: E.g. $\frac{1}{2} = \frac{2}{4}$, yet $1 \neq 2$ and $2 \neq 4$. Equality of fractions only guarantees the cross-multiplied products match: $ad = bc$.

Investigate further: Under what extra condition on a, b, c, d would $\frac{a}{b} = \frac{c}{d}$ actually force $a = c, b = d$?

Section B

1. A student attempts to prove: “For all integers n , if n is odd, then n^2 is odd.”

Line 1: Suppose n is odd, so $n = 2k + 1$ for some integer k .

Line 2: Then $n^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1$.

Line 3: Since $2k^2 + 2k$ is an integer, n^2 is odd.

Is this proof valid? Justify your answer.

Answer: Yes, fully valid.

Method: Every line follows correctly: the algebra in Line 2 is correct $((2k + 1)^2 = 4k^2 + 4k + 1)$, and Line 3 correctly identifies the result as $2 \times (\text{integer}) + 1$, the definition of odd. There is no flaw.

Investigate further: Is this a direct proof or a proof by contrapositive of some related statement?

2. Find the flaw: “Claim: If $x^2 = 3x$, then $x = 3$.” Proof: Divide both sides of $x^2 = 3x$ by x to get $x = 3$.

Answer: Dividing by x is invalid when $x = 0$.

Method: $x^2 = 3x \iff x(x - 3) = 0 \iff x = 0$ or $x = 3$. Dividing by x silently assumes $x \neq 0$ and loses the solution $x = 0$, which also satisfies the original equation.

Investigate further: Rewrite the claim correctly, including both solutions.

3. Find the flaw: “Claim: The only real solution to $\sqrt{x + 7} = x - 5$ is $x = 9$.” Proof: Squaring both sides gives $x + 7 = x^2 - 10x + 25$, so $x^2 - 11x + 18 = 0$, so $(x - 9)(x - 2) = 0$, giving $x = 9$ or $x = 2$.

Answer: $x = 2$ is an extraneous root introduced by squaring; it must be checked against and rejected from the original equation.

Method: Checking $x = 2$ in the original equation: $\sqrt{2 + 7} = \sqrt{9} = 3$, but $x - 5 = 2 - 5 = -3$. Since $3 \neq -3$, $x = 2$ does not satisfy the original equation. Squaring introduced this extraneous solution; only $x = 9$ (which does check: $\sqrt{16} = 4 = 9 - 5$) is valid.

Investigate further: Why does squaring specifically introduce solutions where the two sides had opposite sign?

4. A student argues: For every integer n , $6\left(\frac{4n + 1}{2} - \frac{n - 3}{2}\right)$ is a multiple of 3. (*after TMUA 2020 Paper 2 Q3*)

Line 1: $6\left(\frac{4n + 1}{2} - \frac{n - 3}{2}\right) = 3\left(2\left(\frac{4n + 1}{2} - \frac{n - 3}{2}\right)\right)$.

Line 2: $= 3(4n + 1 - n - 3)$.

Line 3: $= 3(3n - 2)$.

Line 4: which is always a multiple of 3.

On which line does the argument first contain an error, and what exactly is that error?

Answer: Line 2

Method: Correctly distributing, $2\left(\frac{4n + 1}{2} - \frac{n - 3}{2}\right) = (4n + 1) - (n - 3) = 4n + 1 - n + 3 = 3n + 4$, not $4n + 1 - n - 3$ as written. The sign of -3 was not flipped when distributing the subtraction across the bracket $(n - 3)$. (Interestingly, since Line 1’s outer factor of 3 is untouched by this error, the final conclusion — a multiple of 3 — remains true regardless; the argument is still invalid even though its

conclusion happens to be correct.)

Investigate further: Compute the genuinely correct simplified form, $3(3n + 4)$, and confirm it is still a multiple of 3.

5. Find the flaw: “Claim: For every integer n , $n^2 + n + 2$ is even.” Proof: If n is even, $n = 2k$, so $n^2 + n + 2 = 4k^2 + 2k + 2 = 2(2k^2 + k + 1)$, which is even. Hence $n^2 + n + 2$ is even for every integer n .

Answer: The proof only checks the even case; the odd case is never addressed.

Method: To prove a claim for “every integer n ”, both parities must be checked. (The claim does happen to also hold for odd n : $n = 2k + 1 \implies n^2 + n + 2 = 4k^2 + 4k + 1 + 2k + 1 + 2 = 4k^2 + 6k + 4 = 2(2k^2 + 3k + 2)$, even — but the proof as given never establishes this.)

Investigate further: Complete the missing odd case explicitly.

6. Find the flaw: “Claim: For real x , $x^2 = 9 \iff x = 3$.” Proof: If $x = 3$, then $x^2 = 9$. This proves the biconditional.

Answer: Only the $\Leftarrow$ direction ($x = 3 \implies x^2 = 9$) was shown; the $\implies$ direction ($x^2 = 9 \implies x = 3$) is false, since $x = -3$ also satisfies $x^2 = 9$.

Method: A biconditional requires both directions to be proved. The correct statement is $x^2 = 9 \iff x = 3$ or $x = -3$.

Investigate further: What is the correct biconditional relating $x^2 = 9$ to a single equation in x ?

7. Find the flaw: “Claim: For every positive integer n , the n -th Fibonacci number F_n (with $F_1 = F_2 = 1$) is either equal to 1 or prime.” Proof: $F_3 = 2$, $F_4 = 3$, $F_5 = 5$ — all prime. Since the pattern holds for $n = 3, 4, 5$, the claim is proved. What is the smallest counterexample?

Answer: Checking finitely many cases does not prove a universal claim; $F_6 = 8 = 2^3$ is the smallest counterexample.

Method: $F_1 = 1, F_2 = 1, F_3 = 2, F_4 = 3, F_5 = 5, F_6 = 8$. $F_6 = 8$ is neither equal to 1 nor prime ($8 = 2 \times 4$), disproving the claim. Three verified cases give no logical guarantee about the fourth.

Investigate further: Is F_n prime for infinitely many n ? This remains an open problem in general.

8. A student wants to prove: “If n is not divisible by 3, then n^2 is not divisible by 3.” Instead, they show: “If n is divisible by 3, then n^2 is divisible by 3.” Explain why this does not prove the original claim.

Answer: The statement shown is neither the original claim nor its contrapositive — it proves nothing about the original.

Method: Original claim: $\neg A \implies \neg B$ where A : “ $3 \mid n$ ”, B : “ $3 \mid n^2$ ”. Its valid contrapositive is $B \implies A$ (“if $3 \mid n^2$ then $3 \mid n$ ”). The student instead proved $A \implies B$, a different (though also true) statement, which establishes neither the original claim nor its contrapositive.

Investigate further: State the actual contrapositive of the original claim, and outline how you would prove it.

9. Find the flaw: “Claim: For all integers a, b , if a and b are both even, then $a + b$ is even.” Proof:

Since $a + b$ is even, $a + b = 2m$ for some integer m . Since a, b are even, $a = 2j, b = 2k$, so $2j + 2k = 2m$, i.e. $j + k = m$, which is true since m is an integer. Hence $a + b$ is even.

Answer: Circular reasoning: the proof begins by assuming $a + b$ is even (the very thing to be proved) instead of deriving it.

Method: A correct proof starts from $a = 2j, b = 2k$ and derives $a + b = 2j + 2k = 2(j + k)$, which is even because $j + k$ is an integer — it never needs to assume $a + b = 2m$ up front.

Investigate further: Rewrite this as a correct, non-circular direct proof.

10. A student is told: “It is not the case that every student in the class passed the exam.” They conclude: “Therefore, every student in the class failed the exam.” What is wrong with this conclusion?

Answer: The correct negation of “every student passed” is “at least one student did not pass”, not “every student failed”.

Method: $\neg(\forall x P(x)) \equiv \exists x \neg P(x)$, not $\forall x \neg P(x)$. Some students may have passed while at least one did not.

Investigate further: Which named error (from Section A of an earlier drill) does this conclusion resemble?

Section C

1. Consider the claim: “For every positive integer n , $2n^2 + 2n + 1$ is prime.” (after TMUA 2022 Paper 2 Q7)

Answer: D

Method: Lines I–III are correct: the discriminant of $2x^2 + 2x + 1$ is indeed -4 , so the polynomial has no real (let alone integer linear) factorisation. But Line IV commits a category error: a polynomial being irreducible over $\mathbb{Z}[x]$ says nothing about whether its integer OUTPUT VALUES are prime. At $n = 3$: $2(9) + 6 + 1 = 25 = 5^2$, composite, disproving the original claim.

Investigate further: Find another value of n for which $2n^2 + 2n + 1$ is composite.

2. Let P : “ $x^2 - 5x + 6 = 0$ ” and Q : “ $x = 2$ ”. A student argues $Q \implies P$ is true and concludes $P \iff Q$.

Answer: B

Method: $x^2 - 5x + 6 = (x - 2)(x - 3) = 0$, so P is satisfied by $x = 2$ or $x = 3$. $Q \implies P$ is indeed true, but $P \implies Q$ is false since $x = 3$ satisfies P without satisfying Q . Asserting $P \implies Q$ merely because the statements “seem related” is the flaw.

Investigate further: What is the correct relationship (necessary/sufficient) between P and Q ?

3. A student attempts to prove: “For every integer n , $n^2 + n$ is divisible by 2”, checking only the even case.

Answer: B

Method: $n^2 + n = n(n + 1)$ is a product of two consecutive integers, so it is even regardless of the parity of n — but the given proof only demonstrates this for even n and never addresses odd n , so as written it is incomplete.

Investigate further: Complete the missing odd case: for $n = 2k + 1$, show $n^2 + n$ is even.

4. A student attempts to prove: “No three consecutive odd positive integers can all be prime.”
(after TMUA 2023 Paper 2 Q4)

Answer: C

Method: Lines I and II are correct: among $n - 2, n, n + 2$, exactly one is always divisible by 3. Line III is the flaw — being a positive multiple of 3 only implies compositeness if the multiple is *greater than* 3; the multiple could equal 3 itself, which is prime. Exactly this happens at $n = 5$: the three consecutive odd integers 3, 5, 7 are all prime.

Investigate further: Are there any other counterexamples to the original claim besides 3, 5, 7?

5. A student “proves” $\forall n \exists m (m > n)$ and concludes $\exists M \forall n (M > n)$.

Answer: B

Method: Swapping the order of a universal and existential quantifier is not a valid logical step in general. $\forall n \exists m (m > n)$ says every n has SOME larger m (true, e.g. $m = n + 1$); $\exists M \forall n (M > n)$ says a SINGLE M exceeds every n simultaneously, which is false (no integer is larger than every integer).

Investigate further: Give a general example of a true $\forall \exists$ statement whose $\exists \forall$ swap is false.

6. A student “confirms” the converse of “rhombus $\implies$ perpendicular diagonals” using only the example of a square.

Answer: B

Method: One example (the square) supports the converse but cannot prove a universal statement. A kite has perpendicular diagonals but is generally not a rhombus, disproving the converse.

Investigate further: Sketch a kite that is not a rhombus, and confirm its diagonals are perpendicular.

7. A student disproves (*): “For every pair of real numbers x, y , $|x + y| < |x| + |y|$ ” using the pair $x = -2, y = -5$. (after TMUA 2020 Paper 2 Q8)

Answer: D

Method: This is a trap: the student’s reasoning is actually completely correct. (*) as stated uses strict $<$, and at $x = -2, y = -5$ (both negative, hence same sign), $|x + y| = 7 = |x| + |y|$, so $7 < 7$ is false — a fully valid single counterexample to a universal claim. No further pairs need be checked to reject (*).

Investigate further: For which pairs (x, y) does $|x + y| = |x| + |y|$ hold with equality?

8. A student “verifies” “four equal sides $\iff$ square” by checking only that every square has four equal sides.

Answer: B

Method: This checks only necessity (square $\implies$ four equal sides). Sufficiency (four equal sides $\implies$ square) is false: a non-square rhombus has four equal sides but is not a square.

Investigate further: What additional condition, combined with “four equal sides”, would give a genuine necessary and sufficient characterisation of a square?

Section D

1. A flawed attempt to prove: if $s^2 \geq 4p$ then there exist real u, v with $u + v = s, uv = p$. (after TMUA 2021 Paper 2 Q11)

Answer: B

Method: Every algebraic step (II, III) is correct, but the logical direction is backward. The student assumed u, v exist and derived $s^2 \geq 4p$ from that assumption — this proves the CONVERSE of the theorem, not the theorem itself. A correct proof constructs u, v directly: let u, v be the two roots of $t^2 - st + p = 0$; since the discriminant $s^2 - 4p \geq 0$, these roots are real, and by Vieta's formulas $u + v = s, uv = p$.

Investigate further: Why does the discriminant condition $s^2 - 4p \geq 0$ guarantee the roots of $t^2 - st + p = 0$ are real?

2. A student claims “ $n^2 - n + 11$ prime $\implies n^2 + n + 11$ prime” for all positive n , verified for $n = 1, \dots, 9$.

Answer: (a) Finite verification is not a universal proof. (b) $n = 10$.

Method: (a) Checking 9 cases gives no guarantee about $n = 10$ or beyond — a single unchecked value could be a counterexample.

(b) At $n = 10$: $n^2 - n + 11 = 100 - 10 + 11 = 101$, which is prime (premise true). But $n^2 + n + 11 = 100 + 10 + 11 = 121 = 11^2$, which is NOT prime (conclusion false). So the implication fails at $n = 10$, the smallest such value since $n = 1, \dots, 9$ were verified to hold.

Investigate further: Verify that $n^2 - n + 11$ is prime for $n = 1, \dots, 9$ by computing the first few values.

3. Evaluate a contradiction-style proof that “ ab even $\implies a$ even or b even”.

Answer: D) I and II only

Method: The algebra $(2j + 1)(2k + 1) = 4jk + 2j + 2k + 1 = 2(2jk + j + k) + 1$ is correct, so III is false. Statement I is true: assuming the negation (a, b both odd) and deriving a contradiction (ab odd, contradicting ab even) is a valid contradiction proof. Statement II is also true: the derivation “ a, b both odd $\implies ab$ odd” is exactly the contrapositive of the original claim (contrapositive of ab even $\implies (a$ even or b even) is $\neg(a$ even or b even) $\implies \neg(ab$ even), i.e. “ a, b both odd $\implies ab$ odd”), proved here directly without needing the “suppose for contradiction” framing at all.

Investigate further: Rewrite this proof in pure contrapositive form, removing the contradiction framing entirely.

4. Evaluate the “every odd prime > 3 has the form $6k \pm 1$ ” argument.

Answer: B

Method: Every line is correct: $6k, 6k + 2, 6k + 4$ are even (Line 2, correctly dismissed); $6k + 3 = 3(2k + 1)$ is a multiple of 3 exceeding 3 whenever $k \geq 1$, and the case $k = 0$ giving exactly 3 is excluded by the claim's own hypothesis “greater than 3” (Line 3, correctly handled). The proof only ever claimed to show that primes > 3 MUST have the form $6k \pm 1$ (necessity) — it never claimed every number of that form IS prime (sufficiency), and the original claim, read carefully, only asserts the former. So there is no actual gap.

Investigate further: Give an example of a number of the form $6k + 1$ that is not prime, to confirm sufficiency genuinely fails.

5. Evaluate: “For all $n \geq 2$, $n^2 - 1$ is composite”, proved via $n^2 - 1 = (n - 1)(n + 1)$.

Answer: (a) Line 3. (b) $n = 2$; and $n = 2$ is the unique exception.

Method: (a) Line 3 is the gap: a product of two integers each “at least 1” is not necessarily composite — compositeness requires BOTH factors to be at least 2. Line 2 only guarantees $n - 1 \geq 1$, not $n - 1 \geq 2$. (b) At $n = 2$: $n^2 - 1 = 3 = 1 \times 3$. The factor of 1 does not make 3 composite — 3 is prime, so the conclusion genuinely fails here. *Proof that $n = 2$ is the only failure:* for every $n \geq 3$, $n - 1 \geq 2$ and $n + 1 \geq 4$, so $n^2 - 1 = (n - 1)(n + 1)$ is a product of two integers both at least 2, which is precisely the definition of composite — no case analysis or further checking is needed once $n \geq 3$, since the factorisation itself, together with $n - 1 \geq 2$, is a fully general and complete argument covering every such n simultaneously.

Investigate further: Why does the same style of argument fail to show $n^2 - 4$ is composite for all $n \geq 3$? (Check $n = 3$: $n^2 - 4 = 5$.)

“A wrong proof that reads well is far more dangerous than one that does not. Today’s all read well.”

Found a mistake or want to contribute a sheet?

Visit the open-source project at github.com/The-CerealDev/speed-maths